

side PCB for the circuit of the ATmega8515 development board (Fig. 5) is shown in Fig. 7 and its component layout in Fig. 8. Assemble the circuit on a PCB to minimise time and assembly errors. Carefully assemble the components and double-check for any overlooked error.

Downloading your test code

Once simulation is complete, you may want to see its working on your own development board. To program the ATmega8515, connect the 6-wire cable between ISP connectors of the STK500 board (marked as ISP6PIN) and the target board (marked as ISP CON).

The STK500 is a complete starter kit and development system for the AVR Flash microcontroller from Atmel Corporation. It is designed to give designers a quick start to develop code on the AVR and for prototyping and testing of new designs. The

STK500 is supported by AVR Studio. The information on STK500 and other AVR tools can be found in the AVR section of the Atmel website (www.atmel.com).

Connect a serial cable from the connector marked 'RS232 CTRL' on the STK500 board to a COM port on the PC. A three-pin connector (JP) is provided on development board. Connect the reset pin of the ISP connector to reset pin 9 of ATmega8515 microcontroller by shorting the jumper (JP) to 'M' point. M and S are meant for 'master' and 'slave' mode respectively for serial data transfer application.

Now start AVR Studio 4.0 without opening any project file. Optionally, you may proceed as follows: Main Menu→Tools→Program AVR→Select AVR Programmer. Press 'Connect...' after selecting STK500 or AVRISP in the platform window and COM port (say, COM1) in the port window.

After selection, 'AVR Studio

STK500 Programming Menu' window appears as shown in Fig. 6. Select ATmega8515 as 'Device,' ISP as 'Programming Mode' and browse your project hex file from Project→Debug→Exe folder.

Press 'Program' button on the STK500 dialogue box to burn the hex file into your microcontroller. Now remove the JP jumper from 'M' point. Disconnect your device from the ISP cable and press reset switch. Now you can use an LED at port PA0 of ATMEGA8515 to check whether it flashes once every second. A current-limiting resistor (470 ohm) in series with the LED may be connected between PA0 and ground to check blinking of the LED.

EFY note. The source code of this article has been included in this month's EFY-CD and is also available on efymag.com website. ●

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